

Project Report

HR Analytics - Predict Employee Attrition

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Date: October 18, 2025 | Internship: Elevate Labs

Introduction

Employee attrition is a critical challenge that leads to significant operational costs and productivity loss. This project aims to identify the key drivers of employee turnover by analyzing the IBM HR dataset. The core objective is to build a predictive model to identify at-risk employees and create an interactive dashboard to visualize these insights, enabling proactive retention strategies.

Abstract

This project involved an end-to-end data analysis of a fictional IBM HR dataset. Using Python (pandas, scikit-learn), the data was cleaned, prepared, and used to train a Logistic Regression model to predict employee attrition, achieving **88.32% accuracy**. Key findings were then visualized in a dynamic Microsoft Power BI dashboard. The analysis confirmed that job role, monthly income, and department are significant factors in employee turnover.

Tools Used

- Programming Language:** Python 3 (with pandas & scikit-learn libraries)
- Business Intelligence Tool:** Microsoft Power BI (with DAX)
- Development Environment:** Jupyter Notebook

Steps Involved in Building the Project

1. **Data Cleaning:** Loaded the dataset in Python, removed irrelevant columns, and converted the `Attrition` target variable to a binary (1/0) format.
2. **Modeling:** Trained a Logistic Regression model on a 75/25 train-test split of the data after preparing categorical features.
3. **Evaluation:** Tested the model's performance, achieving an overall accuracy of 88.32% and generating a classification report.
4. **Visualization:** Imported the cleaned data into Power BI, created DAX measures for key metrics, and built an interactive dashboard with charts and slicers to explore attrition drivers.

Conclusion

This project successfully translated raw HR data into actionable intelligence. The model can flag at-risk employees, and the dashboard clearly communicates the "why" to stakeholders. The analysis revealed that job roles (specifically Sales Representatives), low monthly income, and department are major factors in attrition. Based on these findings, it is recommended to review compensation for high-risk roles, focus on engagement for early-career employees, and conduct targeted exit interviews to gather more qualitative data.